Unveiling Strategic Nodes in Global Food Trade

Project Milestone - Group 8

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Project Presentation

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Background: Supply Chain Risks

The World is Connected, but Fragile

- **Real-world Threat:** Blockades in the Strait of Hormuz.
- **Impact:** Stops global shipping of food and fertilizer.
- **The Big Picture:** The global food system relies on a few "Strategic Nodes". We need to find them.

Motivation: The Flaw in Traditional HITS

Why do we need Geographic Bias?

- **Traditional HITS Algorithm:** Great for ranking "In" (Import) and "Out" (Export) trade flows.
- **The Flaw:** It only looks at trade volume. It ignores physical distance.
- **Our Solution:** Real-world trade happens in physical space. We add a **Geographic Bias** to fix this flaw and reveal true trade power.

Literature Review

- **Trade & Distance:** Chaney (2014) - Distance acts as a friction in trade.
- **Network Roles:** Kleinberg (1999) - HITS algorithm for "Hubs" and "Authorities".
- **System Attack:** Albert (2000) - Targeted attacks on hubs can collapse a network.

Data Pipeline: Global 2024 Dataset

Source: FAOSTAT Detailed Trade Matrix (1.98 million raw records)

Cleaning the Noise:

1. **Single Countries:** No regions (like "EU" or "World").
2. **Physical Weight:** Use 'Tonnes' for survival crops (Wheat, Rice, etc.).
3. **Tonnage Threshold:** Remove tiny trades (below 25th percentile).

Result: A clean matrix with 189 countries and 2,289 edges.

Data Pipeline: Smart Splitting

Challenge: Trade data has a "heavy-tail" distribution.

Solution: Rank-Stratified Split

- We balanced the training and test sets by trade weight.
- **Fact:** Our 20% test set captures **20.5%** of global trade tonnes.
- This ensures our model evaluation is fair and realistic.

Mathematical Framework: HITS Algorithm

We identify two key roles for each country:

- **Authority (a):** Major sources of food (The "Sinks").
- **Hub (h):** Major distributors/exporters (The "Sources").

How it works:

$$a_i = \sum_{j \to i} h_j \quad \text{and} \quad h_i = \sum_{i \to j} a_j$$

High-quality Hubs point to high-quality Authorities, and vice versa.

Mathematical Framework: Geographic Bias

Logic: Penalize Distance (Gravity Model)
Longer distances mean higher transport costs and more trade resistance.

Updated Weight (W'):

$$W'_{ij} = \frac{W_{ij}}{(d_{ij} + 1)^{\gamma}}$$

- W_{ij} = Original trade weight.
- d_{ij} = Great-Circle Distance between countries.
- γ = Gravity parameter (Distance friction).

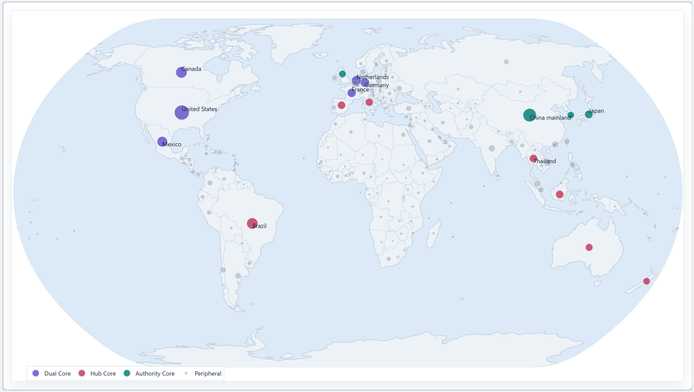
Initial Findings: 2021 European Baseline

Testing our code on 197 global nodes (European Exports).

Scatter Plot: Clear stratification of Hub/Authority roles.

Initial Findings: Global Scale (2021)

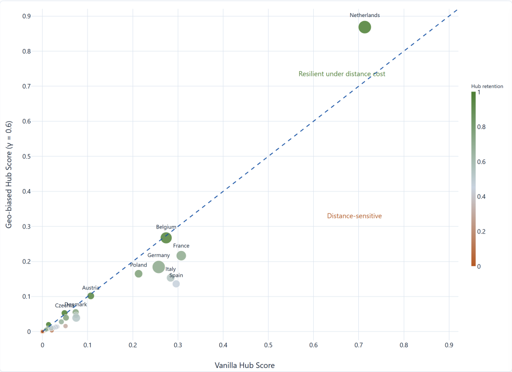
Expanding the HITS analysis to the full global dataset.



Visualizing trade flows across all continents.

Initial Findings: Applying Geographic Bias

Updating weights based on the Gravity Formula.



MOST RESILIENT HUBS

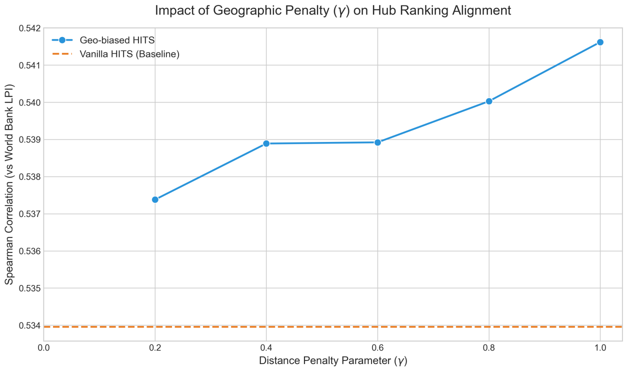
#1	**Netherlands**	122%
#2	**Belgium**	98%
#3	**France**	70%
#4	**Germany**	72%
#5	**Poland**	78%
#6	**Italy**	54%
#7	**Spain**	46%

Identifying nodes that remain strong despite distance costs.

Evaluation I: External Alignment (Validation)

Validating with World Bank LPI

- **Metric:** Spearman Rank Correlation vs. **Logistics Performance Index (LPI)**.
  - A World Bank survey scoring logistics "friendliness" (infrastructure, customs efficiency, international shipments etc.).
- **Hypothesis:** Strategic Hubs should correlate with superior logistics.
- **Key Finding:** As γ increases, correlation with LPI improves, proving Geo-biased HITS identifies **real physical hubs** better than Vanilla HITS.

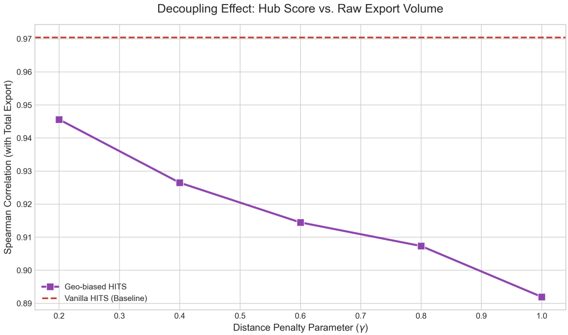


Correlation improves as geographic penalty γ increases.

Evaluation II: Decoupling from Raw Volume

Does the model just find "Big Exporters"?

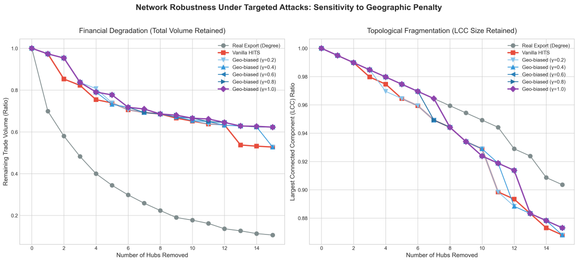
- **Vanilla HITS:** Correlation with Out-degree ≈ 0.97 (Redundant).
- **Geo-biased HITS:** Decouples Hub scores from raw trade volume.
- **Result:** It reveals "Hidden Hubs" that aren't the largest producers but are the most **connected** geographically.



Evaluation III: Robustness & Cascading Failure

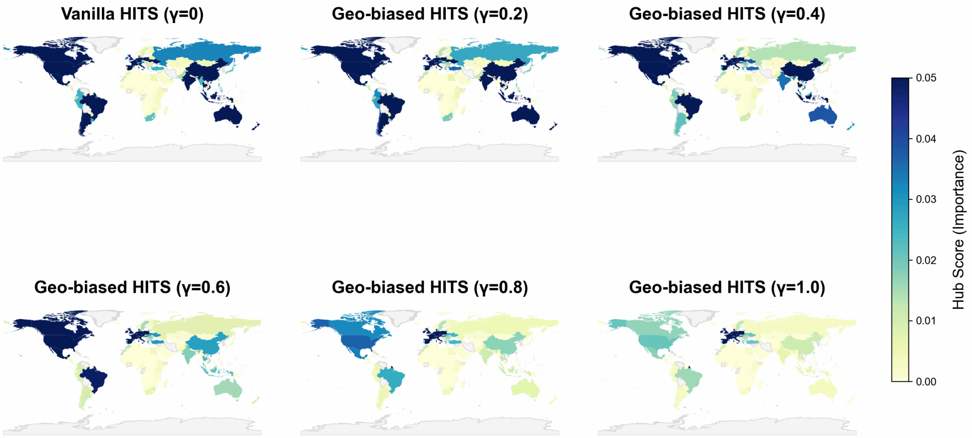
Targeted Attack Simulation

- **Method:** Sequentially remove Top-K nodes and measure Network Fragmentation (LCC).
- **Finding:** Removing Geo-biased Hubs collapses the global network slightly **slower** than removing Vanilla Hubs.
- **Conclusion:** This could due to the fact that we penalize the distance instead of rewarding.



Lower LCC ratio indicates higher system vulnerability.

Visualization: Global Hub Migration ($\gamma = 0 \rightarrow \gamma = 1$)



The shift from "Resource Giants" (Americas) to "Geographic Hubs" (Europe/Eurasia).

Future Work

- **Temporal Analysis:** Extending the model to a 10-year period to observe hub stability during global crises (COVID-19, Conflicts).
- **Redesign bias:** Instead of penalizing distance, we will reward long-range links to discover "Global Anchors" scattered across the world.

$$W' = W \cdot f(d)$$

where $f(d)$ increases with distance. Shift focus from "Local Clusters" to "Strategic Inter-continental Bridges".

Thank You!

Questions & Discussion